

SRINIDHI SHETTY

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Education

Dayananda Sagar College of Engineering

Bangalore, Karnataka

Bachelor of Engineering – Electronics & Communication Engineering

Aug 2023 – June 2027

- CGPA: 7.16 / 10.0
- Relevant Coursework: Digital System Design, Computer Architecture, VLSI Design, Microprocessors & Microcontrollers, Signals & Systems, Embedded Systems, Data Structures & Algorithms, Computer Organization, CMOS Technology.

Experience

IEEE Bangalore – DSCE Chapter

Bangalore, Karnataka

Active Member – Technical Committee

2024 – Present

- Actively contributed to technical events, workshops, and seminars focused on electronics, embedded systems, and emerging hardware technologies.
- Collaborated with peers on IEEE-driven hardware hackathons and innovation challenges, gaining hands-on experience in system-level design and performance analysis.

Extracurricular: IEDC (Entrepreneurship Club), Theatre Club, Literature Club.

Projects

Verilog-Based Implementation of AES Encryption | *SystemVerilog, ModelSim, Vivado*

- Designed and implemented a fully functional **AES-128 encryption engine** in Verilog, covering SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
- Performed **RTL simulation and functional verification** using ModelSim, validating correctness against standard NIST test vectors.
- Analyzed **throughput, latency, and resource utilization** metrics on Xilinx FPGA using Quartus Prime synthesis reports, optimizing critical path for maximum clock frequency.

Multi Clock Domain Controller IC | *SystemVerilog, Quartus Prime, EDA Playground*

- Designed a **Multi Clock Domain (MCD) Controller** IC to manage safe data transfer across asynchronous clock domains using synchronizer FIFOs and handshaking protocols.
- Implemented **Clock Domain Crossing (CDC)** techniques including two-flop synchronizers and Gray-coded FIFO pointers to eliminate metastability issues.
- Verified timing closure and **latency characteristics** across clock boundaries using functional simulation and timing analysis in Quartus Prime.

Smart Electric Vehicle Energy Network (SEVEN) | Peer-to-Peer EV Energy Sharing Platform

- Designed a smart **EV energy-sharing network** to reduce range anxiety by **enabling peer-to-peer (P2P) energy transfer** and intelligent charging station discovery.
- Developed a workflow **integrating battery health monitoring (SoC/SoH)**, user authentication, **AI-assisted** route optimization, and **energy exchange** management.
- Proposed a secure credit-based transaction system that rewards energy donors and facilitates **seamless energy** sharing between EVs and the **power grid**.
- Researched **V2V/V2G** charging technologies and industry standards to support interoperable, sustainable, and future-ready EV ecosystems.

Drishti- Smart AI enabled Glasses for Visually Impaired | Embedded Systems, Hardware AI integration.

- Designed and implemented a **MVP (Minimum Viable Product) prototype** of the Smart Glasses for the Visually Impaired Glasses.
- Researched on product prototyping and market standards along with the business scaling view of the product.
- Got Shortlisted for the CSR initiative of Samsung Electronics – Samsung Solve for Tomorrow.

Achievements & Awards

National Top 50 (Team Drishti) – Samsung Solve for Tomorrow 2025, hosted by Samsung India under the theme *Environment and Sustainability*.

1st Place (Gold Medal) – Nukkad Natak Competition, IIT Kharagpur Spring Fest 2026, demonstrating cross-disciplinary excellence.

Special Domain Award – Ethicare Hardware Hackathon, hosted by DSCE, for innovative hardware design under ethical computing constraints.

Runners-Up – Electrothon 1.0 Hardware Hackathon, hosted by IEEE DSCE Chapter and VinyanaTech.

Technical Skills

Hardware Description Languages: SystemVerilog, Verilog HDL.

Programming Languages: C++, C, Embedded C.

EDA & Simulation Tools: ModelSim, Vivado, Quartus Prime, Cadence Virtuoso, EDA Playground, MATLAB

Operating Systems & Tools: Linux, tcl Scripting

Domains: RTL Design & Verification, FPGA Prototyping, High-Performance Computing, Embedded Systems, VLSI Design, CMOS, Physical Design.

Publications & Patents

“Low Cost, Vehicle Independent, Driver Calibrated Brake Drag Detection and Alarm System”

Indian Patent Application – Submitted

Invented a cost-effective embedded hardware system for real-time brake drag detection, leveraging sensor fusion and signal processing techniques. Filed as an Indian Patent Publication.